Smart Civic Infrastructure Detection Using AI

ABSTRACT

Urban areas face two major challenges: increasing traffic congestion and inefficient civic issue reporting mechanisms. Traditional traffic control systems operate on fixed signal timing without considering real-time vehicle density, while existing grievance redressal systems often suffer from delays, lack of transparency, and manual processing inefficiencies. To address these challenges, this project proposes **UrbanEye AI – An AI-Based Smart Traffic Density and Civic Issue Management System**.

The proposed system integrates computer vision, machine learning, and digital governance technologies to create a unified urban management platform. For traffic management, real-time vehicle detection and counting are performed using deep learning-based object detection models to estimate traffic density levels such as low, moderate, and heavy congestion. Based on the detected density, intelligent decision support can be provided for adaptive traffic signal control and congestion monitoring.

For civic issue management, the system enables citizens to report infrastructure problems such as potholes, garbage overflow, broken streetlights, and road damage through image-based submissions. Computer vision models are used to automatically detect and classify reported issues, while Natural Language Processing (NLP) techniques assist in categorizing complaints and determining severity. GPS-based geo-tagging ensures accurate location tracking, and complaints are routed to relevant authorities through a structured digital platform. An administrative dashboard provides real-time tracking, analytics, escalation mechanisms, and performance monitoring to enhance transparency and accountability.

By integrating traffic density analysis with AI-driven civic issue detection, the proposed system offers a scalable, intelligent, and citizen-centric approach toward smart city development. The solution enhances urban mobility, improves grievance redressal efficiency, and supports data-driven governance for building cleaner, safer, and smarter cities.